

Question:

An 18-year-old man moves from sea level to an elevation of 2,400 m to train as a skier. The increased requirement for oxygen delivery to tissues at the higher elevation stimulates the synthesis of a renal hormone (erythropoietin), which targets hematopoietic stem cells in the bone marrow. Erythropoietin promotes the survival of early erythroid progenitor cells primarily through which of the following mechanisms? Answer Choices: A. Altered cell-matrix adhesion B. Downregulation of p53 C. Enhanced glucose uptake D. Inhibition of apoptosis

Reasoning + Answer:

<think> The question is asking about the mechanism by which erythropoietin promotes the survival of early erythroid progenitor cells. To answer this question, we need to identify the correct biological process involved in erythropoietin's action. Erythropoietin is known to inhibit apoptosis, which is the process of programmed cell death. This is the primary mechanism by which erythropoietin promotes the survival of early erythroid progenitor cells. Therefore, the correct answer is the option that corresponds to inhibition of apoptosis. </think> <answer> Inhibition of apoptosis </answer>

